

Community Emergency Response App (CERA)

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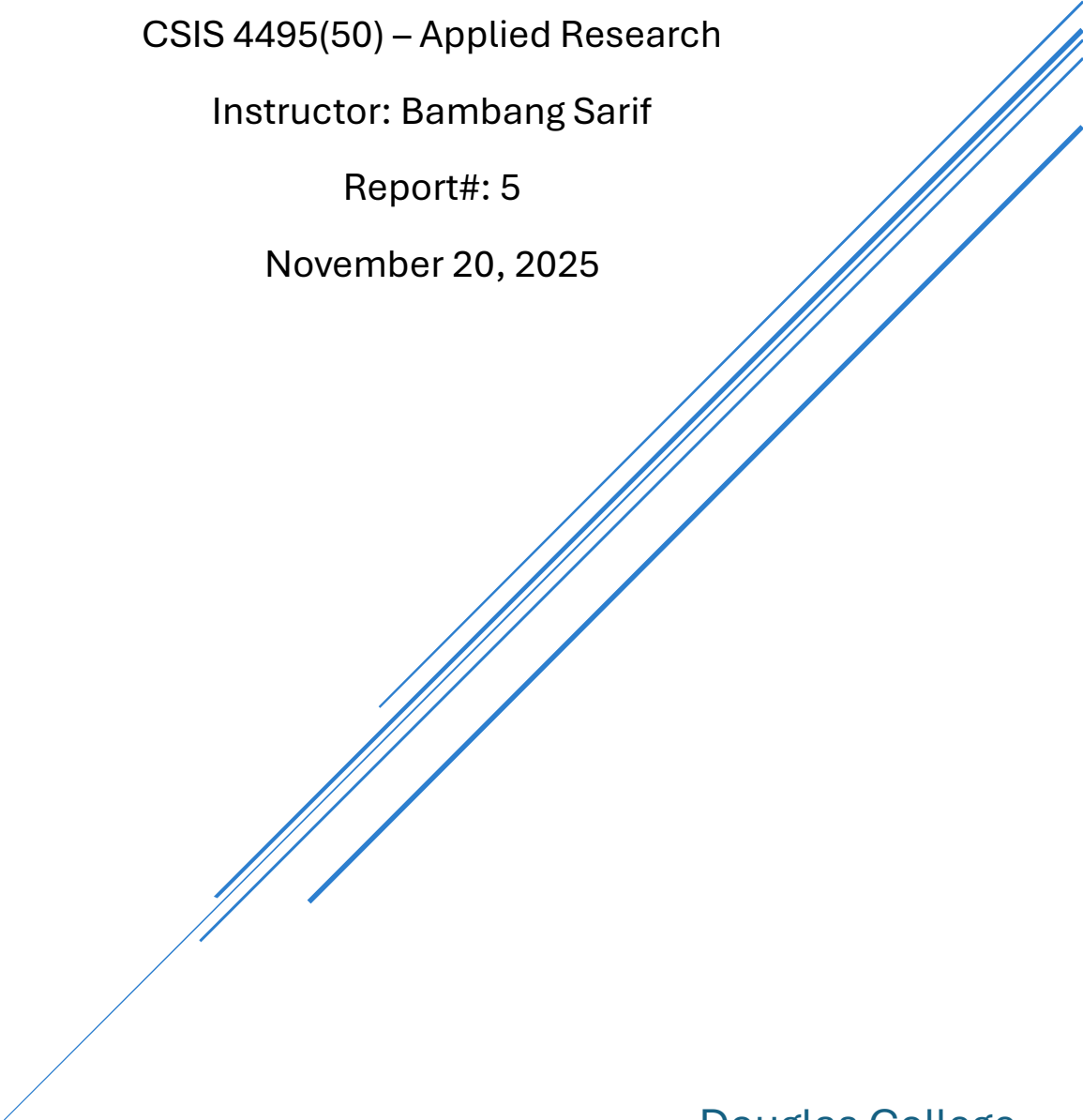
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CSIS 4495(50) – Applied Research

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Report#: 5

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A series of five parallel blue lines of varying lengths, slanted diagonally from the bottom-left towards the top-right, located on the right side of the page.

Douglas College
CSIS 4495

Work Logs

Date	Number Of Hours	Description of work done
Nov 07, 2025	3	Implemented role-based access rules for incidents. Volunteers can view: (1) nearby incidents, and (2) assigned incidents only if accepted . Coordinators can access all. Residents restricted. Added backend checks to enforce this.
Nov 08, 2025	2.5	Fixed volunteer “Failed to load incident” issue by updating backend endpoint /incidents/:id to return incident only if permissions match. Synced frontend logic with new rules.
Nov 10, 2025	4	Rebuilt IncidentDetail UI to include hero banner, gradient backgrounds, severity chips, reporter details, sticky footer button, and dynamic theme support. Added hooks to load incident securely using token.
Nov 12, 2025	3.5	Developed Incident Tracking Timeline Screen with vertical stepper, color-coded actions, timestamps, actor/target identification, and integrated backend logs. Added timeline fetch using secure API.
Nov 14, 2025	3	Added full image gallery system : fullscreen modal, swipeable FlatList carousel, thumbnails, arrow navigation, tap-to-open, and loader for slow images.

		Added responsive scaling for devices.
Nov 15, 2025	2	fixed modal height issues (not filling screen). Adjusted container sizing and overlay to handle notch areas correctly (iPhone notch fix).
Nov 17, 2025	2.5	Improved IncidentCardDetailed → Added skeleton UI, improved assignedVolunteers section, fixed activity log parsing, and improved “reported at” formatting.
Nov 19, 2025	3	Reworked IncidentCard carousel + loader to avoid blocking UI. Added error-handling for broken image URLs + fallback behavior.

Description of Work done

Over the past two weeks, the primary focus was on implementing new functionality for incident tracking and timelines supporting media uploads to improve the real-world usability (as told by professor) of the CERA application.

1. Role-Based Access Control & Incident Security

⇒ Volunteers

- Can view **nearby incidents**
- Can view **assigned incidents ONLY IF accepted**
- Cannot access incidents assigned to others
- Prevented unauthorized access → fixed “Failed to load incident” bug

⇒ Coordinators

- Full access to all incidents
- Can view logs, timeline, reporter info
- Can dispatch volunteers and approve incidents

⇒ Residents

- Can only create reports
- Cannot view nearby incidents
- Cannot open any incident screen

⇒ **Backend Improvements**

- Hardened /incidents/:id endpoint
- Added assignment-based filtering
- Synced role middleware across routes

2. Cloudinary Image Upload Integration

- Removed reliance on local filesystem storage.
- Configured backend to use Cloudinary Node SDK for secure image hosting.
- Implemented multipart/form-data upload from Expo image picker.
- Ensured correct DB storage of returned Cloudinary URLs.
- Added error handling for slow network or failed uploads.

3. Incident Detail Screen Enhancements

⇒ **Implemented:**

- Gradient header bar based on incident type
- Clean two-line location display
- Sticky bottom CTA button ("Contact Coordinator")
- Dynamic color-coded chips
- Better responsive layout
- No notch overlap on iOS
- Improved createdAt formatting

4. Full Image Gallery / Carousel System

- Tap image → fullscreen modal
- Swipeable carousel (FlatList + pagingEnabled)
- Loader until image loads
- Left/right arrow navigation

- Thumbnails at bottom for quick switching
- scrollToIndex sync to active image
- Works with any number of photos
- Handles Cloudinary slow image loads

5. Incident Tracking Timeline

⇒ Developed a vertical stepper-style timeline that displays:

- assignment
- accepted
- in_progress
- completed
- approved
- resolved
- declined

⇒ **Each entry includes:**

- Developed a vertical stepper-style timeline that displays:
- actor
- action
- target (if any)
- timestamp
- message
- color-coded status

Testing & Debugging

- Validated upload times and image retrieval performance.
- MapView header overlapping with notch
- Missing reporterName field in cards
- Duplicate logs
- Infinite loops in useEffect with distance calculation
- Image loading delays in fullscreen modal
- AllVolunteersScreen merge conflict
- Incidents route missing return statements
- Cloudinary fallback checks

Repo Check-In

Checked-in folders and file since report 3:

1. Backend:

- /backend/routes/incidents.js-
 - Added role-based authorization checks
 - Fixed /incidents/:id returning unauthorized for volunteers
 - Added timeline fetch logic
- /backend/utils/cloudinary.js- Created Cloudinary config helper containing environment-bound credentials, secure URL presets, and reusable upload function
- /backend/models/Incident.js-
 - Added logs array formatting
 - Ensured photos array always normalized
- /backend/controllers/incidentController.js-
 - Assignment validation
 - Volunteer acceptance verification
 - Expanded logs returned

2. Frontend:

- /app/components/IncidentCardDetailed.tsx- ew UI, image modal, timeline preview, better chips, loader, volunteer/coordinator logic

- /app/screens/IncidentTracking.tsx- Complete timeline UI implementation, Actor/target logic, Image support added
- /app/components/ImageGalleryModal.tsx- Extracted full gallery logic.
- /app/screens/MapScreen.tsx Improved permission flow, Fixed clustering bugs

Kanban Board

⇒ Newly Completed Tasks:

- **Full Image Gallery Modal**
- **Incident Tracking / Timeline Screen**
- **Role-Based Visibility Control**
- **Volunteer assignment acceptance rules**
- **Incident Detail UI overhaul**
- **Notch fix for all top headers**

⇒ In Review:

- **Improving map performance**
- **Adding route preview using Google Maps**
- **Coordinators dispatch flow**

Images from GitHub Project

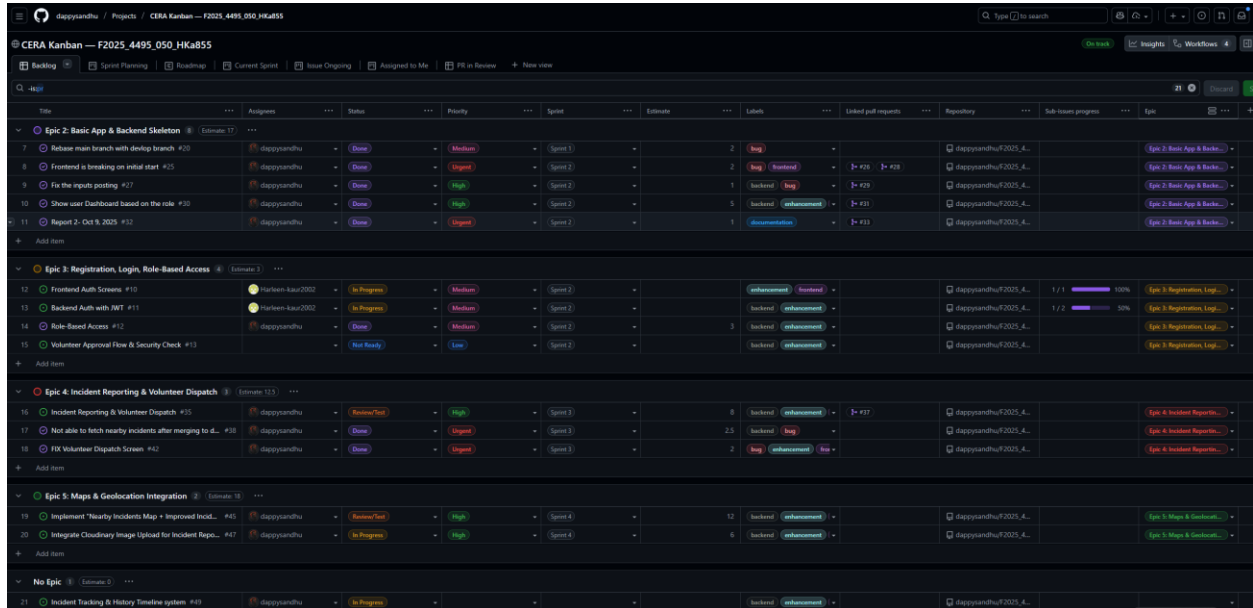


Figure 1: Kanban Board showing tasks in Backlog, In Progress, and Done.

Screens Implemented according to above tasks:

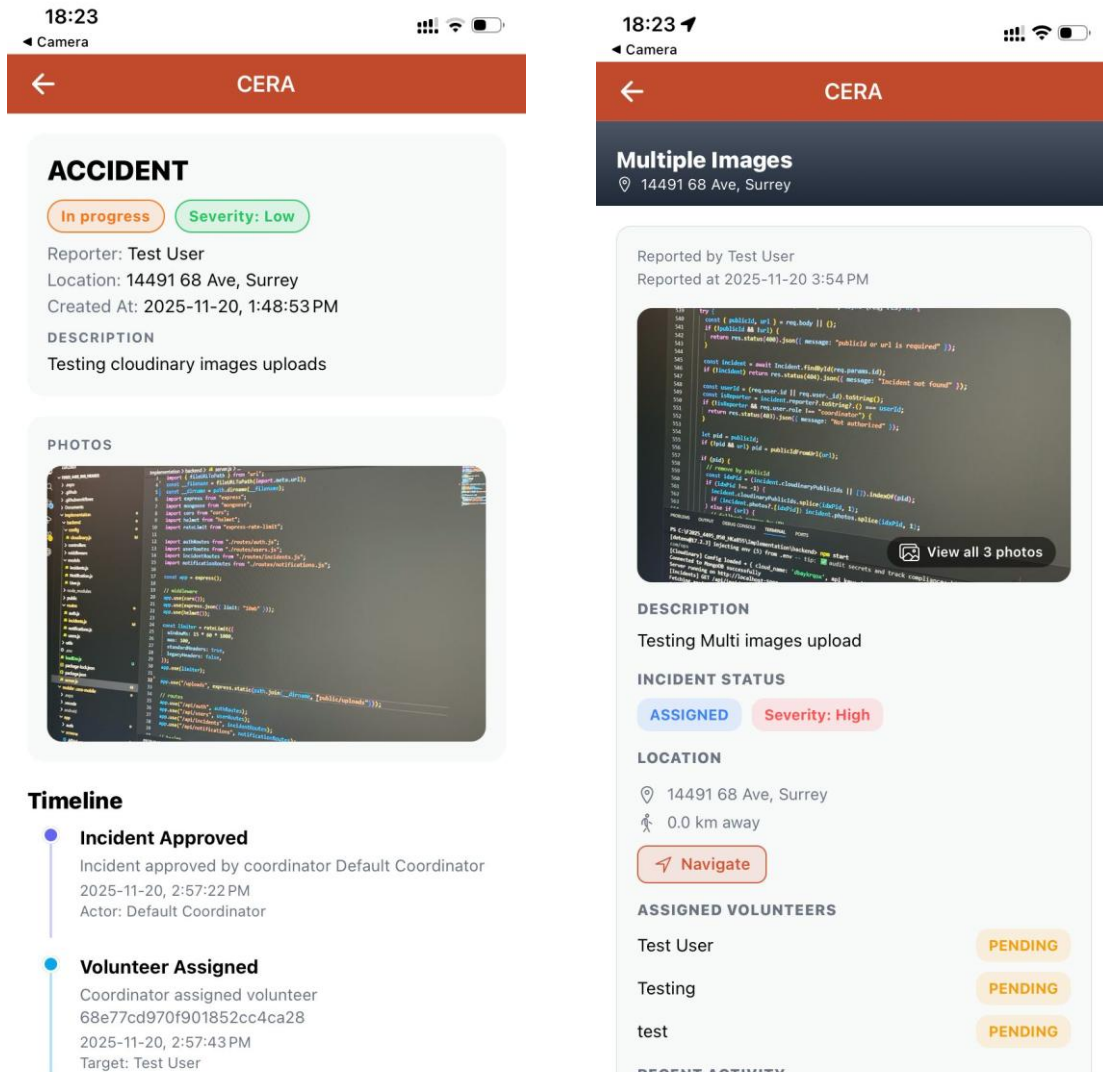


Figure 2. Incident Tracking and Timelines

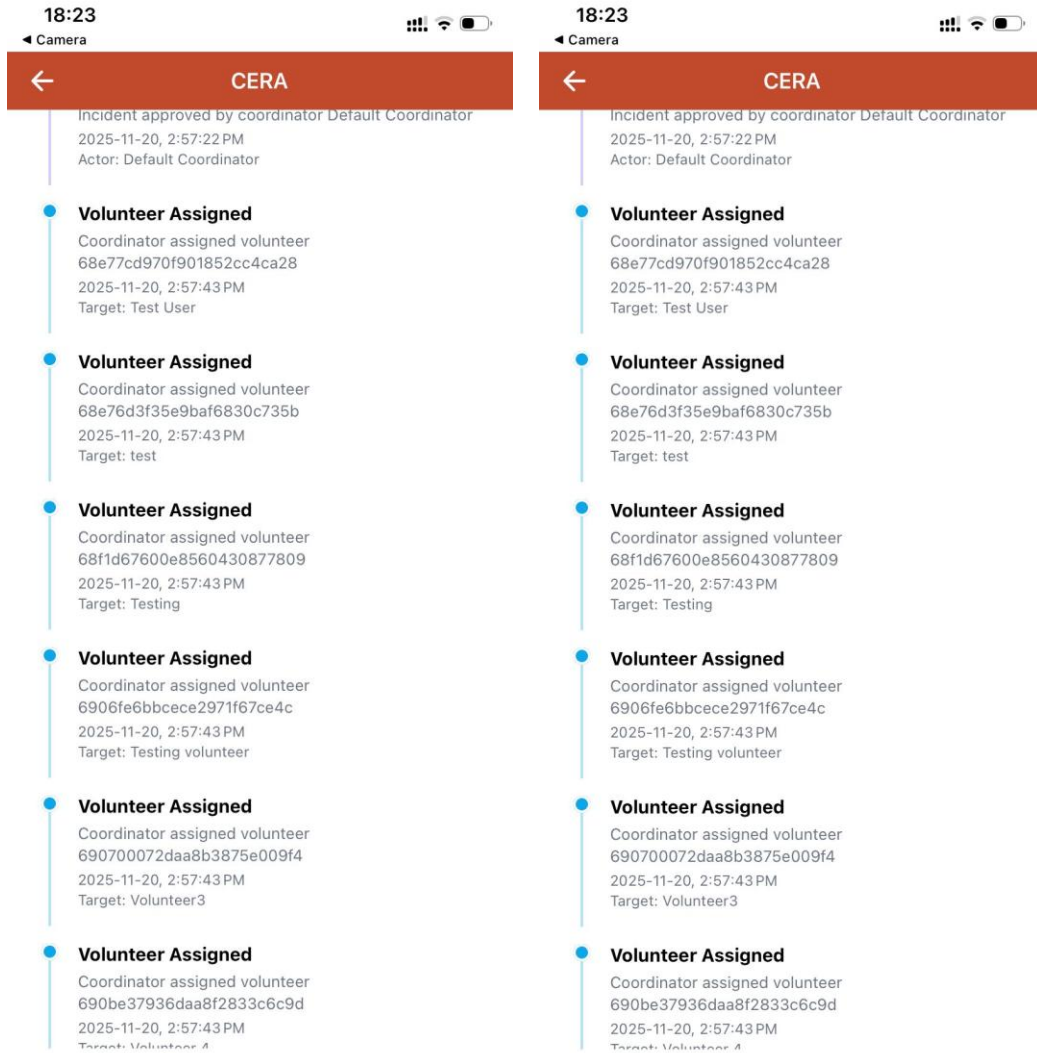


Figure 3: Incident Tracking details

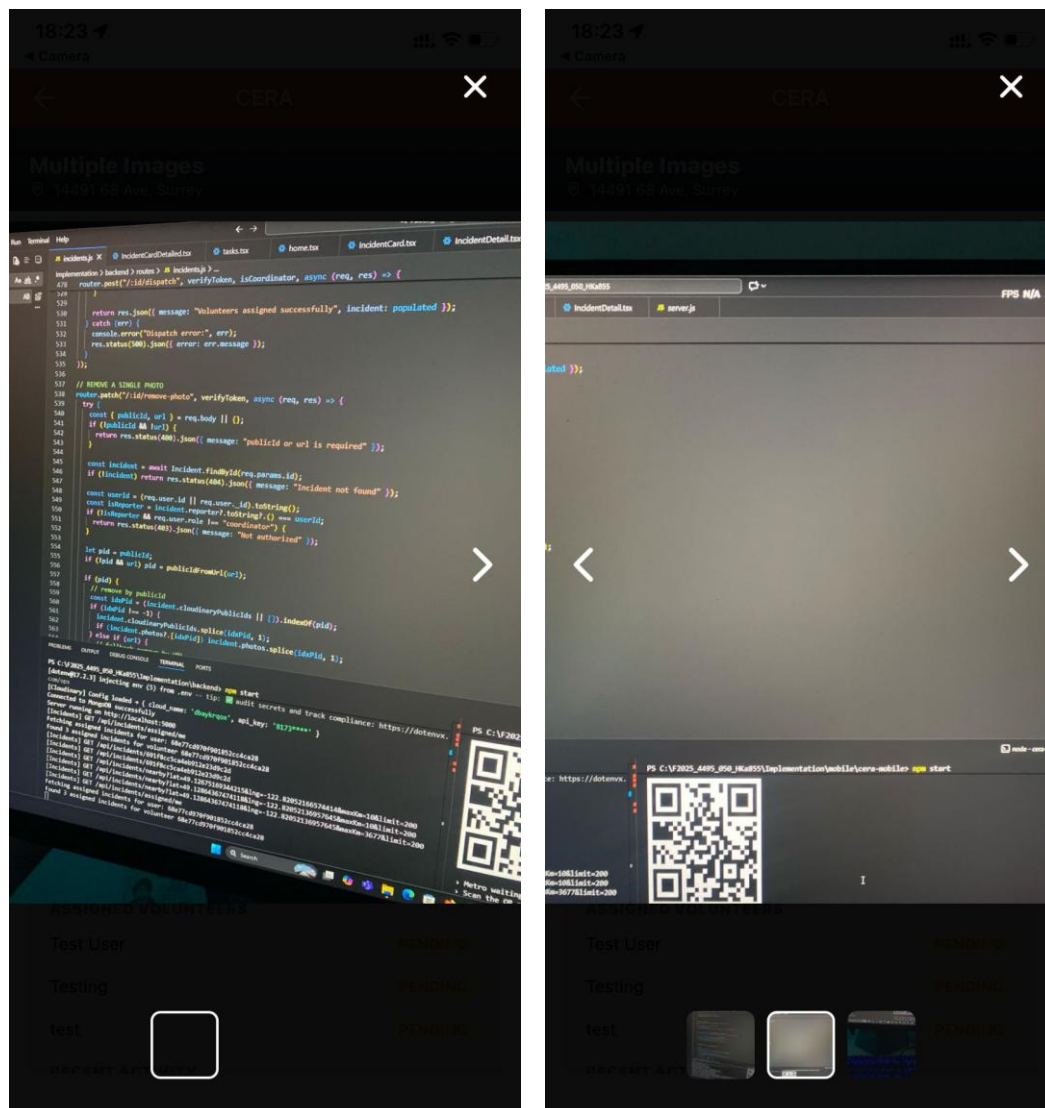


Figure 4: Preview Multiple Images from Cloudinary